

Set 6 Product Sales Analysis Dataset: Monthly Product Sales

Product ID 1 2 3 Product Name Product A Product B

January Sales 2000 February Sales 2200 1500 Product C 1.

1200 1800 1400 March Sales 2400 1600 1100

Create a grouped bar chart to visualize the sales of each product for the first quarter. Label the chart elements. 2. months. 3. 4.

Generate a stacked area chart to represent the overall sales trend for all products over the three Build a table to show the monthly sales data for each product. Label the table elements. Develop a Tableau dashboard combining the grouped bar chart, stacked area chart, and the table for interactive exploration of sales data.

1)

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
import numpy as np
```

```
# Dataset
```

```
data = {  
    "Product ID": [1, 2, 3],  
    "Product Name": ["Product A", "Product B", "Product C"],  
    "January": [2000, 1500, 1200],  
    "February": [2200, 1800, 1400],  
    "March": [2400, 1600, 1100]  
}
```

```
df = pd.DataFrame(data)
print(df)
import matplotlib.pyplot as plt
import numpy as np

products = df["Product Name"]

jan = df["January"]
feb = df["February"]
mar = df["March"]

x = np.arange(len(products))
width = 0.25

plt.figure(figsize=(8,5))

plt.bar(x-width, jan, width, label="January")
plt.bar(x, feb, width, label="February")
plt.bar(x+width, mar, width, label="March")

plt.xticks(x, products)
```

```
plt.title("Monthly Product Sales (First Quarter)")
```

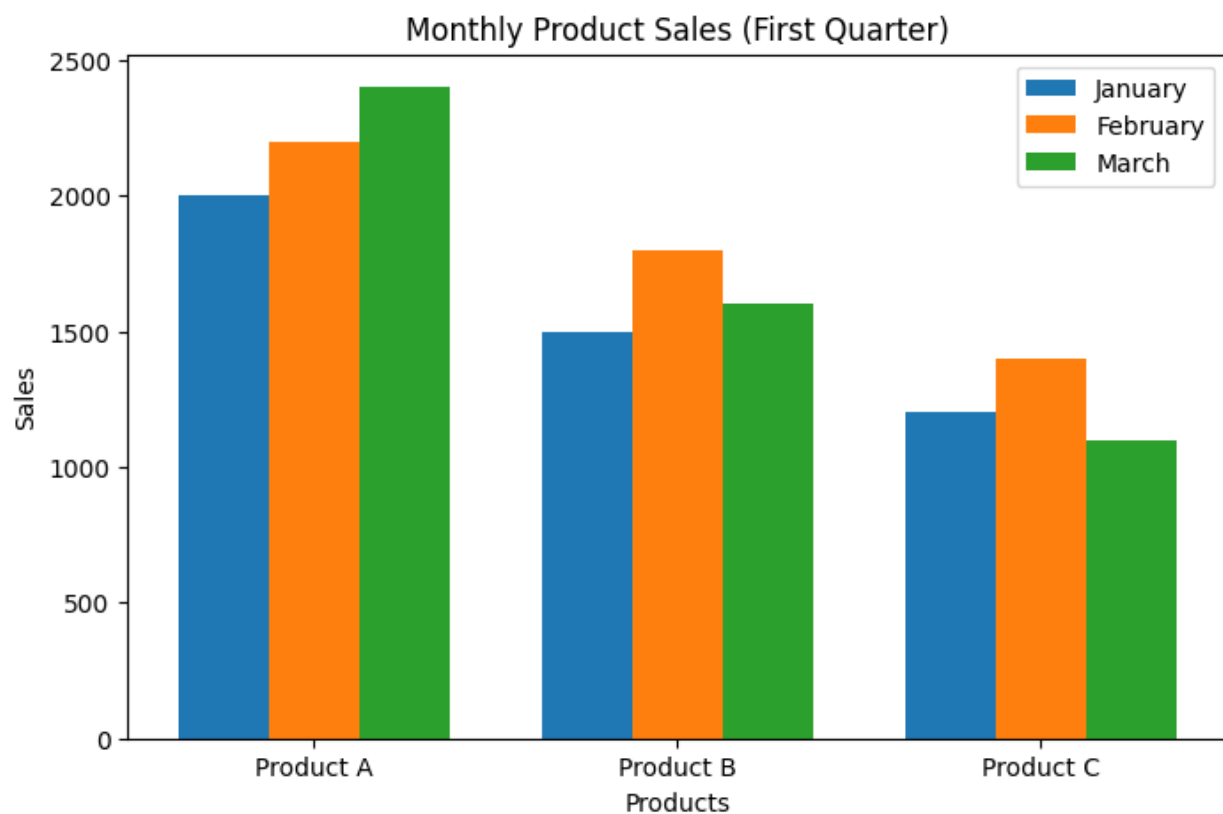
```
plt.xlabel("Products")
```

```
plt.ylabel("Sales")
```

```
plt.legend()
```

```
plt.show()
```

OUTPUT:



b)

```
months = ["January", "February", "March"]
```

```
productA = [2000, 2200, 2400]
```

```
productB = [1500, 1800, 1600]
```

```
productC = [1200,1400,1100]
```

```
plt.figure(figsize=(8,5))
```

```
plt.stackplot(  
    months,  
    productA,  
    productB,  
    productC,  
    labels=["Product A","Product B","Product C"]  
)
```

```
plt.title("Overall Sales Trend")
```

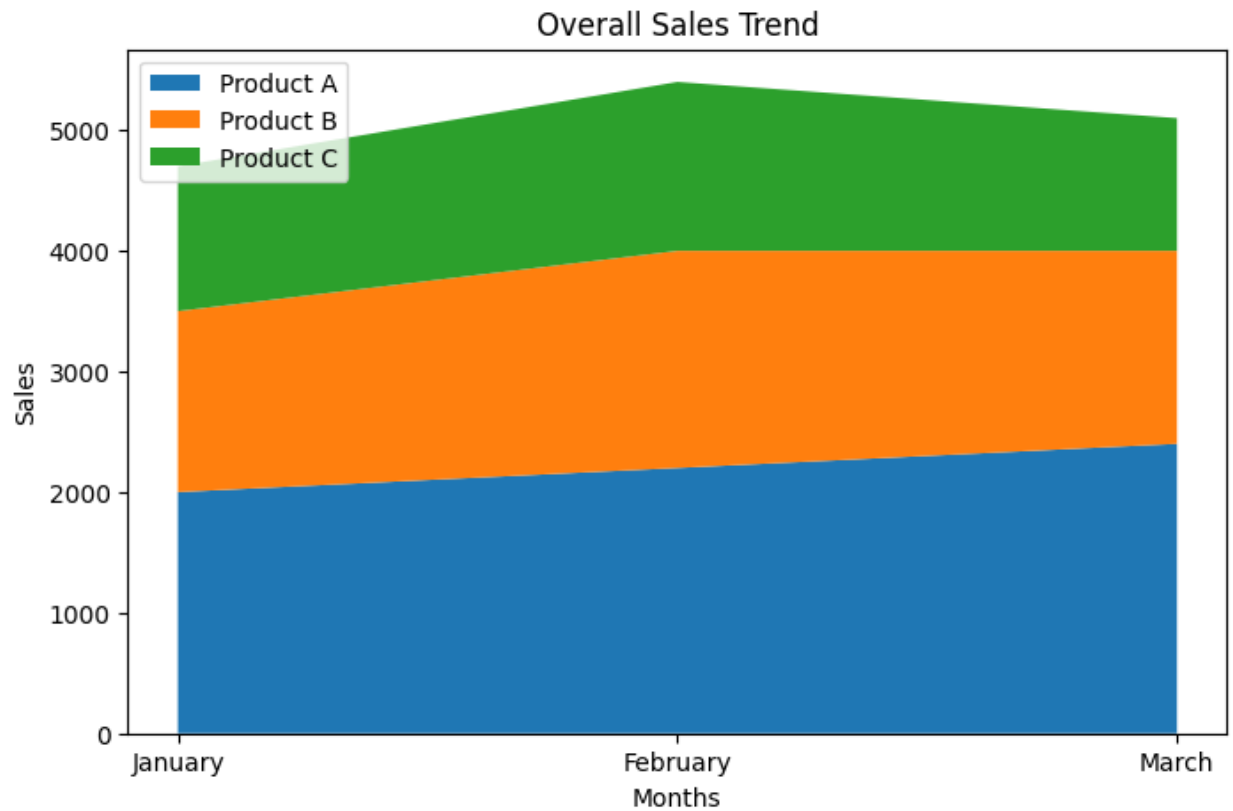
```
plt.xlabel("Months")
```

```
plt.ylabel("Sales")
```

```
plt.legend(loc="upper left")
```

```
plt.show()
```

output:



C)

```
print("\nMonthly Product Sales Table\n")
```

```
print(df.to_string(index=False))
```

output:

Monthly Product Sales Table

	Product ID	Product Name	January	February	March
1	Product A	2000	2200	2400	
2	Product B	1500	1800	1600	
3	Product C	1200	1400	1100	

d)

```
fig = plt.figure(figsize=(14,8))
```

```
#####
```

```
# Grouped Bar Chart
```

```
#####
```

```
ax1 = fig.add_subplot(221)
```

```
x = np.arange(len(products))
```

```
width = 0.25
```

```
ax1.bar(x-width, jan, width, label="January")
```

```
ax1.bar(x, feb, width, label="February")
```

```
ax1.bar(x+width, mar, width, label="March")
```

```
ax1.set_xticks(x)
```

```
ax1.set_xticklabels(products)
```

```
ax1.set_title("Grouped Bar Chart")
```

```
ax1.set_xlabel("Products")
```

```
ax1.set_ylabel("Sales")
```

```
ax1.legend()
```

```
#####

# Stacked Area Chart

#####

ax2 = fig.add_subplot(222)

ax2.stackplot(
    months,
    productA,
    productB,
    productC,
    labels=["Product A","Product B","Product C"]
)

ax2.set_title("Stacked Area Chart")
ax2.set_xlabel("Months")
ax2.set_ylabel("Sales")
ax2.legend()

#####

# Table

#####
```

```
ax3 = fig.add_subplot(212)
```

```
ax3.axis('off')
```

```
table = ax3.table(  
    cellText=df.values,  
    colLabels=df.columns,  
    loc='center'  
)
```

```
table.auto_set_font_size(False)
```

```
table.set_fontsize(10)
```

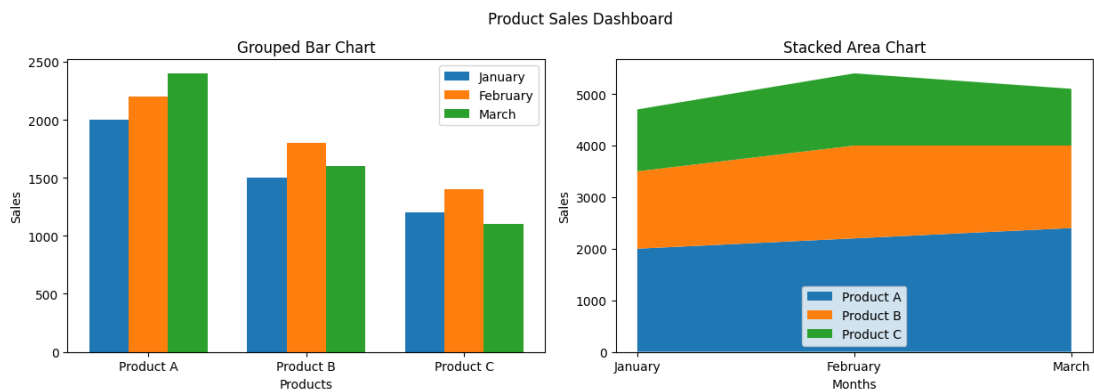
```
table.scale(1.2,2)
```

```
plt.suptitle("Product Sales Dashboard")
```

```
plt.tight_layout()
```

```
plt.show()
```


output:



Product ID	Product Name	January	February	March
1	Product A	2000	2200	2400
2	Product B	1500	1800	1600
3	Product C	1200	1400	1100

Set 7 Customer Demographics Analysis Dataset: Customer Demographics Customer ID 1 2 3 Age 28 Gender Female Income (in \$) 50000 35 42 1. Male Female 60000 75000 Create a bar chart to visualize the distribution of customer ages. Label the axes and title the chart. 2. 3. 4. Generate a pie chart to represent the distribution of customers by gender. Build a table to show the demographic information for each customer. Label the table elements. Develop a Tableau dashboard combining the bar chart, pie chart, and the table for interactive exploration of customer demographics.

A)

```
plt.figure(figsize=(6,5))
```

```
plt.bar(df["Customer ID"].astype(str), df["Age"])
```

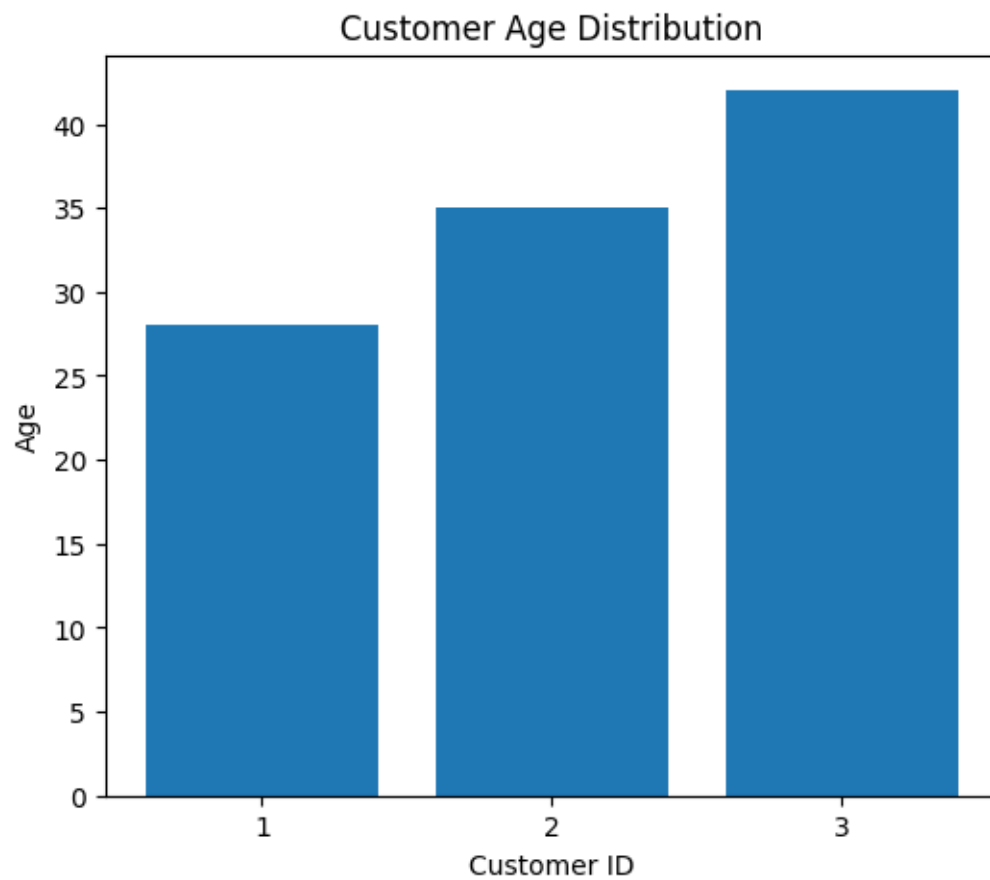
```
plt.title("Customer Age Distribution")
```

```
plt.xlabel("Customer ID")
```

```
plt.ylabel("Age")
```

```
plt.show()
```

output:



b)

```
gender_count = df["Gender"].value_counts()
```

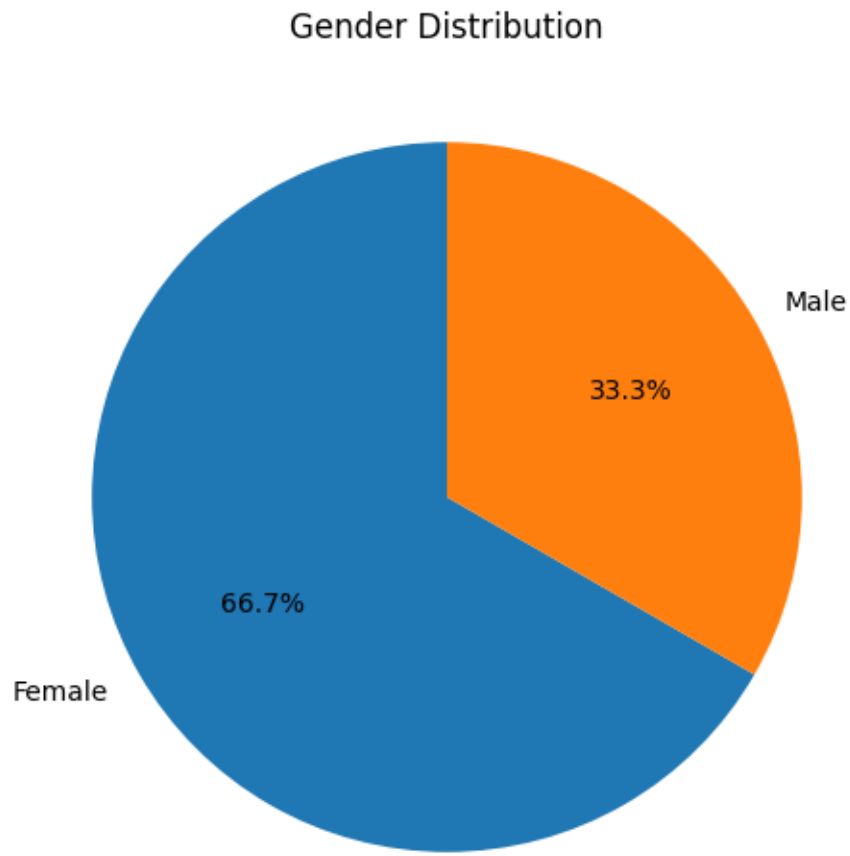
```
plt.figure(figsize=(6,6))
```

```
plt.pie(  
    gender_count,  
    labels=gender_count.index,  
    autopct="%1.1f%%",  
    startangle=90  
)
```

```
plt.title("Gender Distribution")
```

```
plt.show()
```

output:



c)

```
print("\nCustomer Demographics Table\n")
```

```
print(df.to_string(index=False))
```

output:

Customer Demographics Table

Customer ID	Age	Gender	Income
-------------	-----	--------	--------

1	28	Female	50000
---	----	--------	-------

2	35	Male	60000
---	----	------	-------

3 42 Female 75000

d)

```
fig = plt.figure(figsize=(12,8))
```

```
# -----
```

```
# Bar Chart
```

```
# -----
```

```
ax1 = fig.add_subplot(221)
```

```
ax1.bar(df["Customer ID"].astype(str), df["Age"])
```

```
ax1.set_title("Customer Age Distribution")
```

```
ax1.set_xlabel("Customer ID")
```

```
ax1.set_ylabel("Age")
```

```
# -----
```

```
# Pie Chart
```

```
# -----
```

```
ax2 = fig.add_subplot(222)
```

```
gender_count = df["Gender"].value_counts()
```

```
ax2.pie(  
    gender_count,  
    labels=gender_count.index,  
    autopct="%1.1f%%",  
    startangle=90  
)
```

```
ax2.set_title("Gender Distribution")
```

```
# -----
```

```
# Table
```

```
# -----
```

```
ax3 = fig.add_subplot(212)
```

```
ax3.axis("off")
```

```
table = ax3.table(  
    cellText=df.values,  
    colLabels=df.columns,  
    loc="center"  
)
```

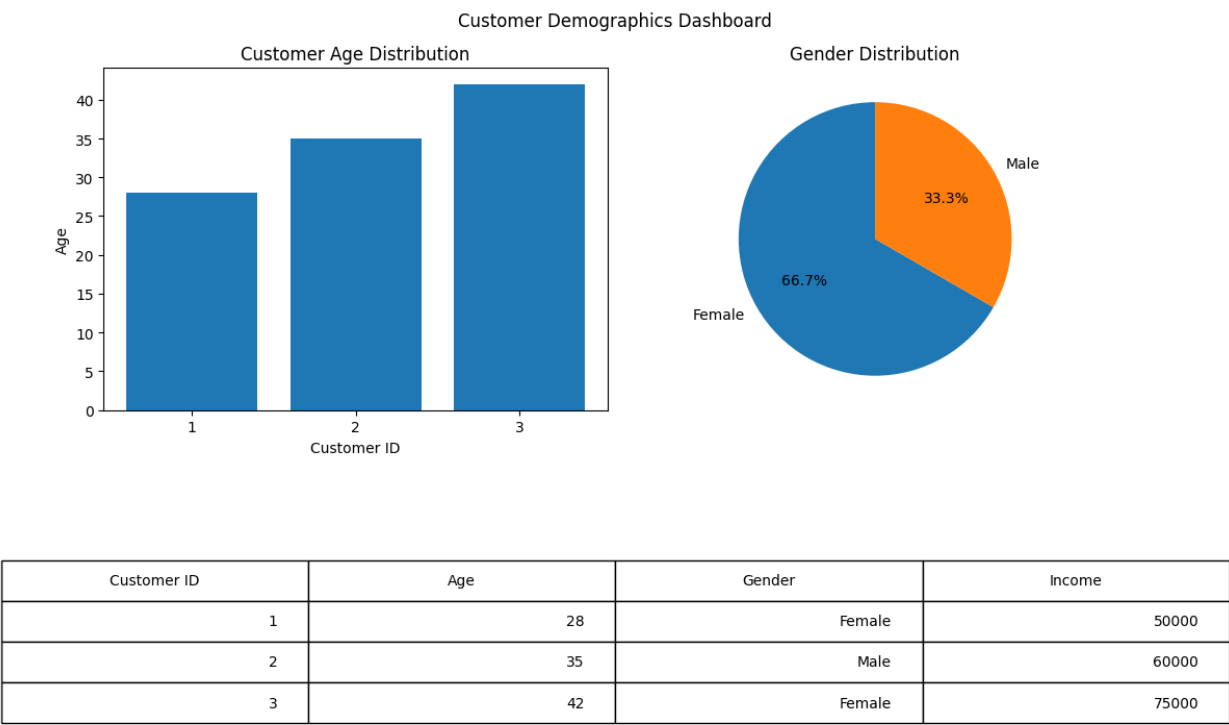
```
table.auto_set_font_size(False)
table.set_fontsize(10)
table.scale(1.2, 2)

plt.suptitle("Customer Demographics Dashboard")

plt.tight_layout()

plt.show()

output:
```



Set 8 Employee Performance Analysis Dataset: Employee Performance

Employee ID	1	2	3	Department	Sales	HR	Years of Service	5	Performance Score	85	3	Marketing	1.7	92	78
-------------	---	---	---	------------	-------	----	------------------	---	-------------------	----	---	-----------	-----	----	----

Create a line chart to visualize the performance trend of employees over time. Label the axes and title the chart. 2. Generate a bar chart showing the distribution of employees across different departments. Label the chart elements. 3. 4. Build a table to display the performance data for each employee. Label the table elements. Develop a Tableau dashboard combining the line chart, bar chart, and the table for interactive exploration of employee performance data.

A)

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
# Dataset
```

```
data = {  
    "Employee ID": [1, 2, 3],  
    "Department": ["Sales", "HR", "Marketing"],  
    "Years of Service": [5, 3, 7],  
    "Performance Score": [85, 92, 78]  
}
```

```
df = pd.DataFrame(data)
```



```
print(df)
```

```
plt.figure(figsize=(6,5))
```

```
plt.plot(  
    df["Employee ID"],  
    df["Performance Score"],  
    marker="o",  
    linewidth=2  
)
```

```
plt.title("Employee Performance Trend")
```

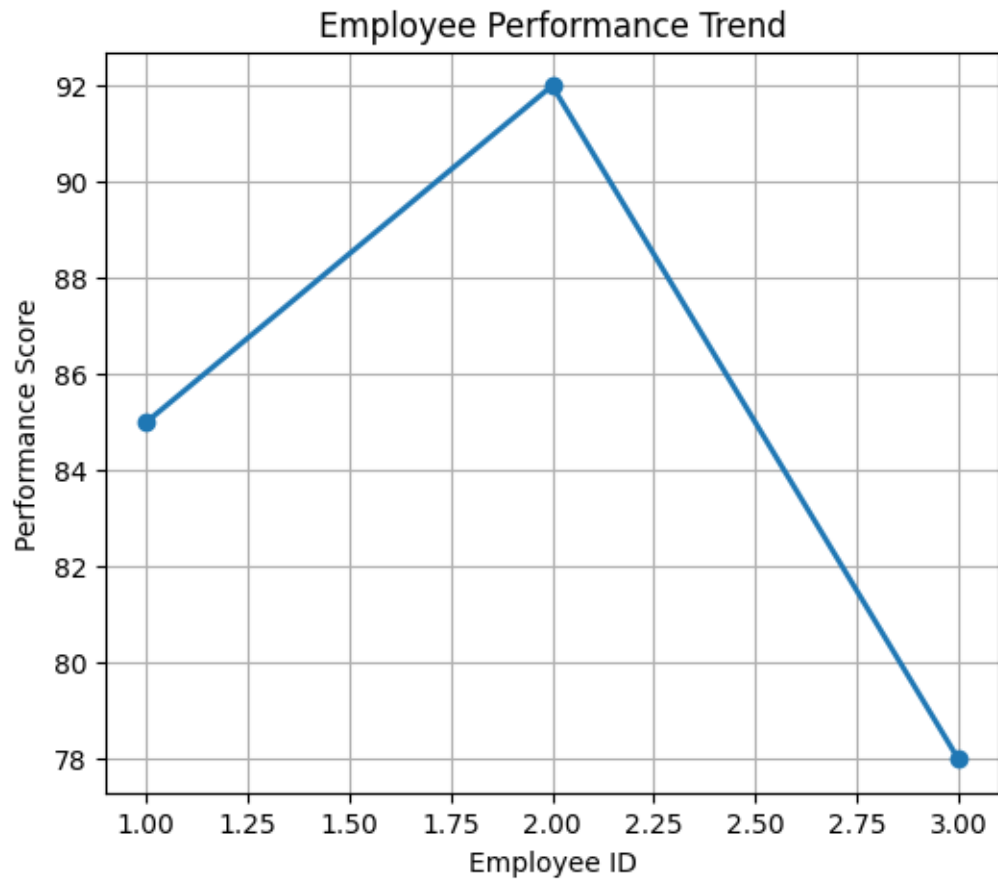
```
plt.xlabel("Employee ID")
```

```
plt.ylabel("Performance Score")
```

```
plt.grid(True)
```

```
plt.show()
```

output:



b)

```
department_count = df["Department"].value_counts()
```

```
plt.figure(figsize=(6,5))
```

```
plt.bar(  
    department_count.index,  
    department_count.values  
)
```

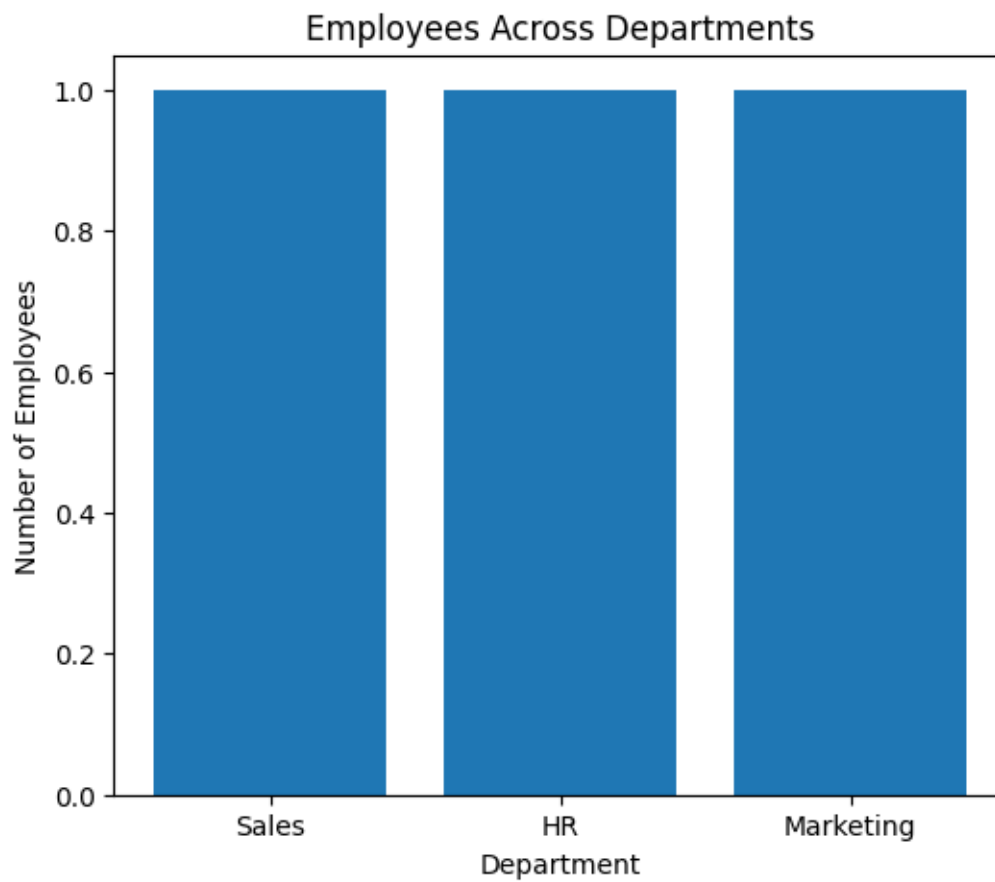
```
plt.title("Employees Across Departments")
```

```
plt.xlabel("Department")
```

```
plt.ylabel("Number of Employees")
```

```
plt.show()
```

output:



C)

```
print("\nEmployee Performance Table\n")
```

```
print(df.to_string(index=False))
```

output:

Employee Performance Table

Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78

d)

```
fig = plt.figure(figsize=(12,8))
```

```
# -----
```

```
# Line Chart
```

```
# -----
```

```
ax1 = fig.add_subplot(221)
```

```
ax1.plot(  
    df["Employee ID"],  
    df["Performance Score"],  
    marker="o",  
    linewidth=2  
)
```

```
ax1.set_title("Employee Performance Trend")
ax1.set_xlabel("Employee ID")
ax1.set_ylabel("Performance Score")
ax1.grid(True)
```

```
# -----
```

```
# Bar Chart
```

```
# -----
```

```
ax2 = fig.add_subplot(222)
```

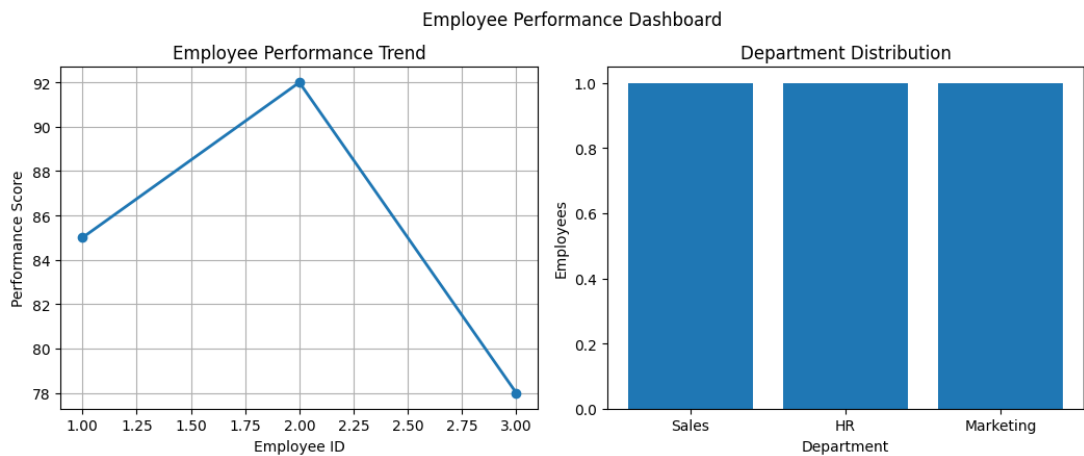
```
department_count = df["Department"].value_counts()
```

```
ax2.bar(
    department_count.index,
    department_count.values
)
```

```
ax2.set_title("Department Distribution")
ax2.set_xlabel("Department")
ax2.set_ylabel("Employees")
```

```
# -----  
# Table  
# -----  
ax3 = fig.add_subplot(212)  
  
ax3.axis("off")  
  
table = ax3.table(  
    cellText=df.values,  
    colLabels=df.columns,  
    loc="center"  
)  
  
table.auto_set_font_size(False)  
table.set_fontsize(10)  
table.scale(1.2, 2)  
  
plt.suptitle("Employee Performance Dashboard")  
  
plt.tight_layout()  
  
plt.show()
```

output:



Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78

**set 9 Dataset: Online_Learning_Activity.csv Student_ID
Gender Age Course Study_Time (hrs) Videos_Watched
Quiz_Score Login_Date L01 Male 20 R 3.5 12 78 L02 L03
L04 L05 L06 Female 22 R Male 19 SQL Female 21 R Male
23 R Female 20 SQL 4.2 2.0 5.0 2.5 4.0 15 8 18 9 14 85 65 92
70 88 2025-01-05 2025-01-05 2025-02-08 2025-02-08 2025-03-
12 2025-03-12**

1. Import the dataset in R. Plot a histogram for Quiz_Score. Draw a boxplot of Quiz_Score by Course. Add title and labels. Interpret student performance.

2. Create a scatter plot using R programming between Study_Time and Quiz_Score. Represent Videos_Watched using bubble size. Apply transparency. Explain the learning

pattern. 3. Using R programming, convert Login_Date into Date format. Calculate average quiz score per month. Plot a line chart. Apply moving average smoothing. Identify the trend. 4. Import Online_Learning_Activity.csv into Tableau. Create a bar chart showing average quiz score by Course. Create a scatter plot (Study Time vs Quiz Score). Add filter for Gender and Course. Design a mini interactive dashboard.

A)

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
# Import Dataset
```

```
import pandas as pd
```

```
data = {
```

```
    "Student_ID": ["L01", "L02", "L03", "L04", "L05", "L06"],
```

```
    "Gender": ["Male", "Female", "Male", "Female", "Male", "Female"],
```

```
    "Age": [20, 22, 19, 21, 23, 20],
```

```
    "Course": ["R", "R", "SQL", "R", "R", "SQL"],
```

```
    "Study_Time (hrs)": [3.5, 4.2, 2.0, 5.0, 2.5, 4.0],
```

```
    "Videos_Watched": [12, 15, 8, 18, 9, 14],
```

```
    "Quiz_Score": [78, 85, 65, 92, 70, 88],
```



```
"Login_Date": ["2025-01-05", "2025-01-05", "2025-02-08", "2025-02-08", "2025-03-12", "2025-03-12"]
```

```
}
```

```
df = pd.DataFrame(data)
```

```
print(df)
```

```
print(df)
```

```
# ----- Histogram -----
```

```
plt.figure(figsize=(6,5))
```

```
plt.hist(df["Quiz_Score"], bins=5, edgecolor="black")
```

```
plt.title("Histogram of Quiz Scores")
```

```
plt.xlabel("Quiz Score")
```

```
plt.ylabel("Number of Students")
```

```
plt.show()
```

```
# ----- Boxplot -----
```

```
plt.figure(figsize=(6,5))
```

```
df.boxplot(column="Quiz_Score", by="Course")
```

```
plt.title("Quiz Score by Course")
```

```
plt.suptitle("")
```

```
plt.xlabel("Course")
```

```
plt.ylabel("Quiz Score")
```

```
plt.show()
```

```
print("\nInterpretation:")
print("Higher quiz scores indicate better student performance.")
print("The boxplot compares quiz score variation across courses.")
```

output:

	Student_ID	Gender	Age	Course	Study_Time (hrs)	Videos_Watched
0	L01	Male	20	R	3.5	12
1	L02	Female	22	R	4.2	15
2	L03	Male	19	SQL	2.0	8
3	L04	Female	21	R	5.0	18
4	L05	Male	23	R	2.5	9
5	L06	Female	20	SQL	4.0	14

	Quiz_Score	Login_Date
0	78	2025-01-05
1	85	2025-01-05
2	65	2025-02-08
3	92	2025-02-08
4	70	2025-03-12
5	88	2025-03-12

	Student_ID	Gender	Age	Course	Study_Time (hrs)	Videos_Watched
--	------------	--------	-----	--------	------------------	----------------

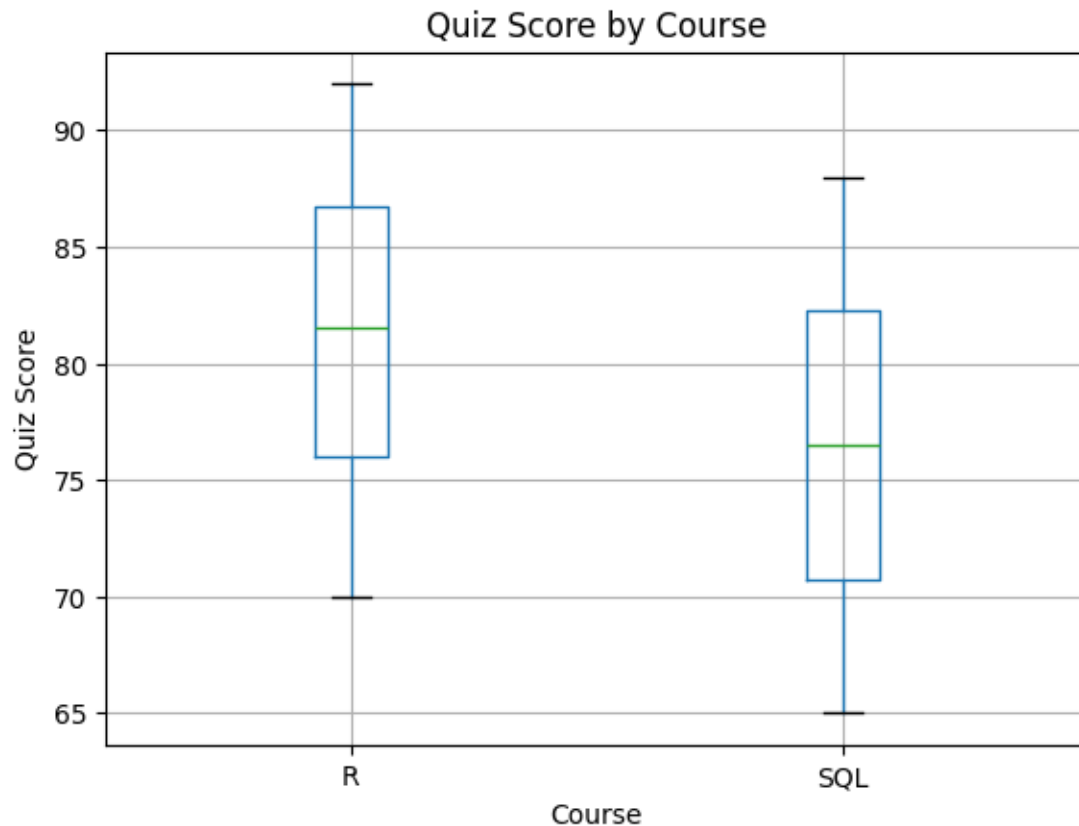
0	L01	Male	20	R	3.5	12
1	L02	Female	22	R	4.2	15
2	L03	Male	19	SQL	2.0	8
3	L04	Female	21	R	5.0	18
4	L05	Male	23	R	2.5	9
5	L06	Female	20	SQL	4.0	14

Quiz_Score Login_Date

0	78	2025-01-05
1	85	2025-01-05
2	65	2025-02-08
3	92	2025-02-08
4	70	2025-03-12
5	88	2025-03-12



<Figure size 600x500 with 0 Axes>



Interpretation:

Higher quiz scores indicate better student performance.

The boxplot compares quiz score variation across courses.

B)

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
plt.figure(figsize=(7,5))
```

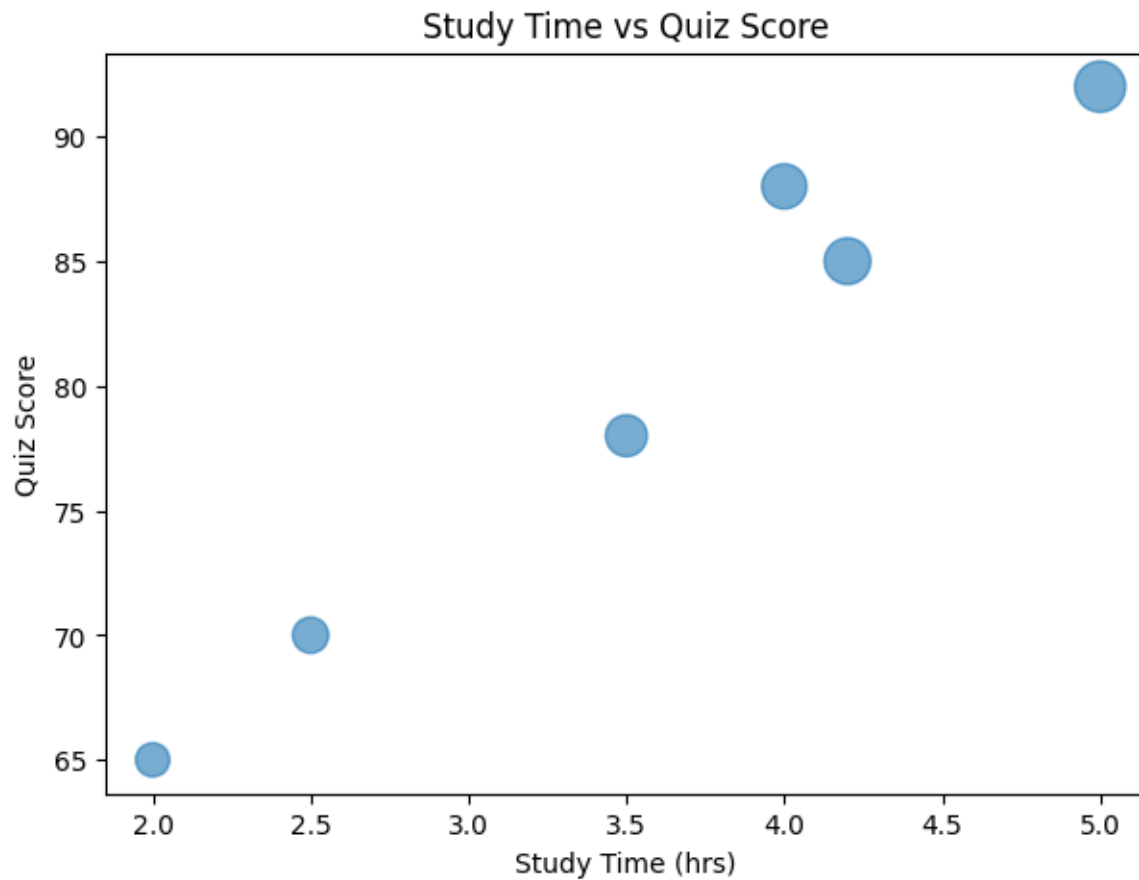
```
plt.scatter(
```

```
    df["Study_Time (hrs)"],
```

```
    df["Quiz_Score"],
```

```
    s=df["Videos_Watched"]*20,
```

```
alpha=0.6
)
plt.title("Study Time vs Quiz Score")
plt.xlabel("Study Time (hrs)")
plt.ylabel("Quiz Score")
plt.show()
print("\nLearning Pattern:")
print("Students with more study time generally score higher.")
print("Larger bubbles represent more videos watched.")
output:
```



Learning Pattern:

Students with more study time generally score higher.

Larger bubbles represent more videos watched.

c)

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
# Convert Login_Date to Date format
```

```
df["Login_Date"] = pd.to_datetime(df["Login_Date"])
```

```
# Extract Month
```

```
df["Month"] = df["Login_Date"].dt.to_period("M")
```

```
# Average Quiz Score per Month
```

```
monthly = df.groupby("Month")["Quiz_Score"].mean()
```

```
# Moving Average
```

```
moving_avg = monthly.rolling(window=2).mean()
```

```
plt.figure(figsize=(7,5))
```

```
plt.plot(monthly.index.astype(str),
```

```
monthly.values,  
marker="o",  
label="Average Score")
```

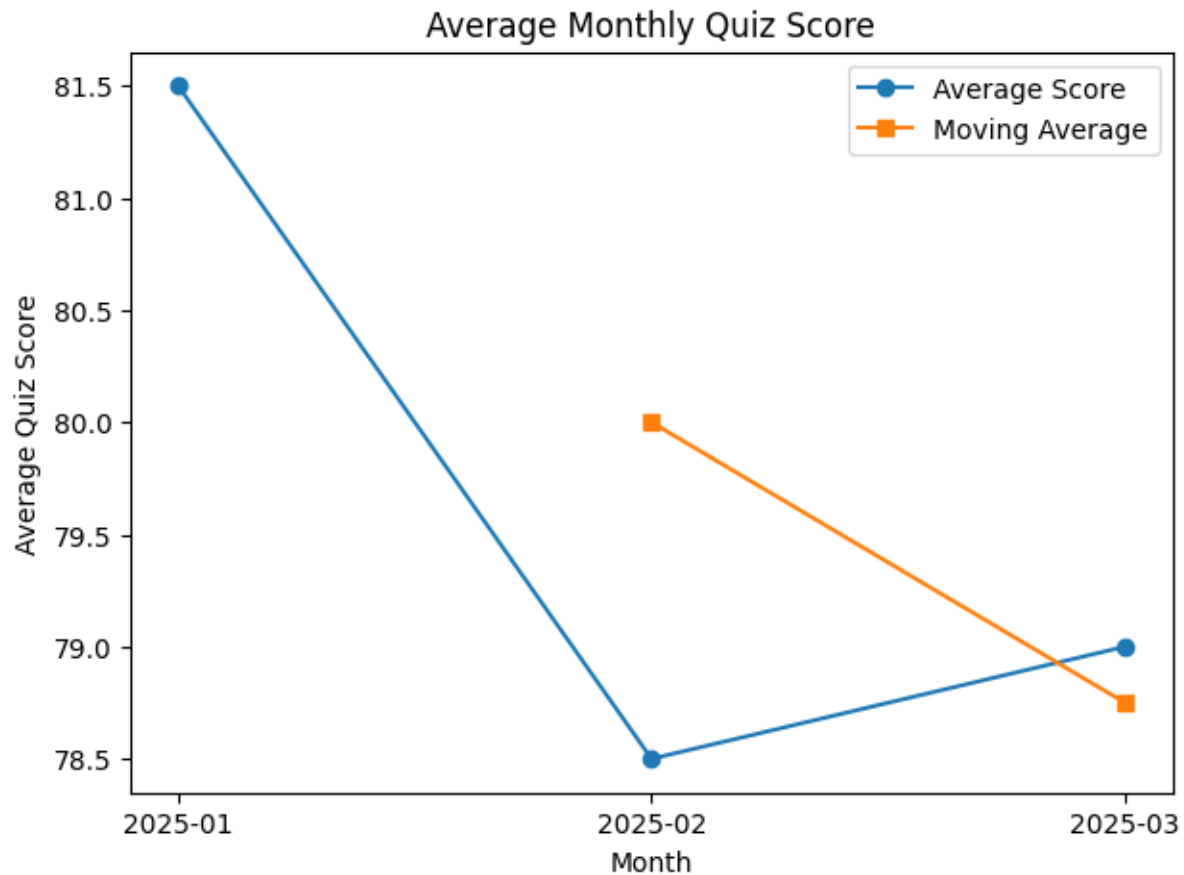
```
plt.plot(moving_avg.index.astype(str),  
moving_avg.values,  
marker="s",  
label="Moving Average")
```

```
plt.title("Average Monthly Quiz Score")  
plt.xlabel("Month")  
plt.ylabel("Average Quiz Score")
```

```
plt.legend()
```

```
plt.show()
```

```
print("\nTrend:")  
print("The line chart shows monthly average quiz scores.")  
print("Moving average smooths short-term fluctuations.")  
output:
```

Trend:

The line chart shows monthly average quiz scores.

Moving average smooths short-term fluctuations.

d)

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
fig = plt.figure(figsize=(12,8))
```

```
# -----
```

```
# Average Quiz Score by Course
# -----

ax1 = fig.add_subplot(221)

course_avg = df.groupby("Course")["Quiz_Score"].mean()

ax1.bar(course_avg.index, course_avg.values)

ax1.set_title("Average Quiz Score by Course")
ax1.set_xlabel("Course")
ax1.set_ylabel("Average Score")

# -----

# Scatter Plot
# -----

ax2 = fig.add_subplot(222)

ax2.scatter(
    df["Study_Time (hrs)"],
    df["Quiz_Score"],
    s=df["Videos_Watched"]*20,
    alpha=0.6
```

```
)
```

```
ax2.set_title("Study Time vs Quiz Score")
```

```
ax2.set_xlabel("Study Time")
```

```
ax2.set_ylabel("Quiz Score")
```

```
# -----
```

```
# Table
```

```
# -----
```

```
ax3 = fig.add_subplot(212)
```

```
ax3.axis("off")
```

```
table = ax3.table(  
    cellText=df.values,  
    colLabels=df.columns,  
    loc="center"
```

```
)
```

```
table.auto_set_font_size(False)
```

```
table.set_fontsize(8)
```

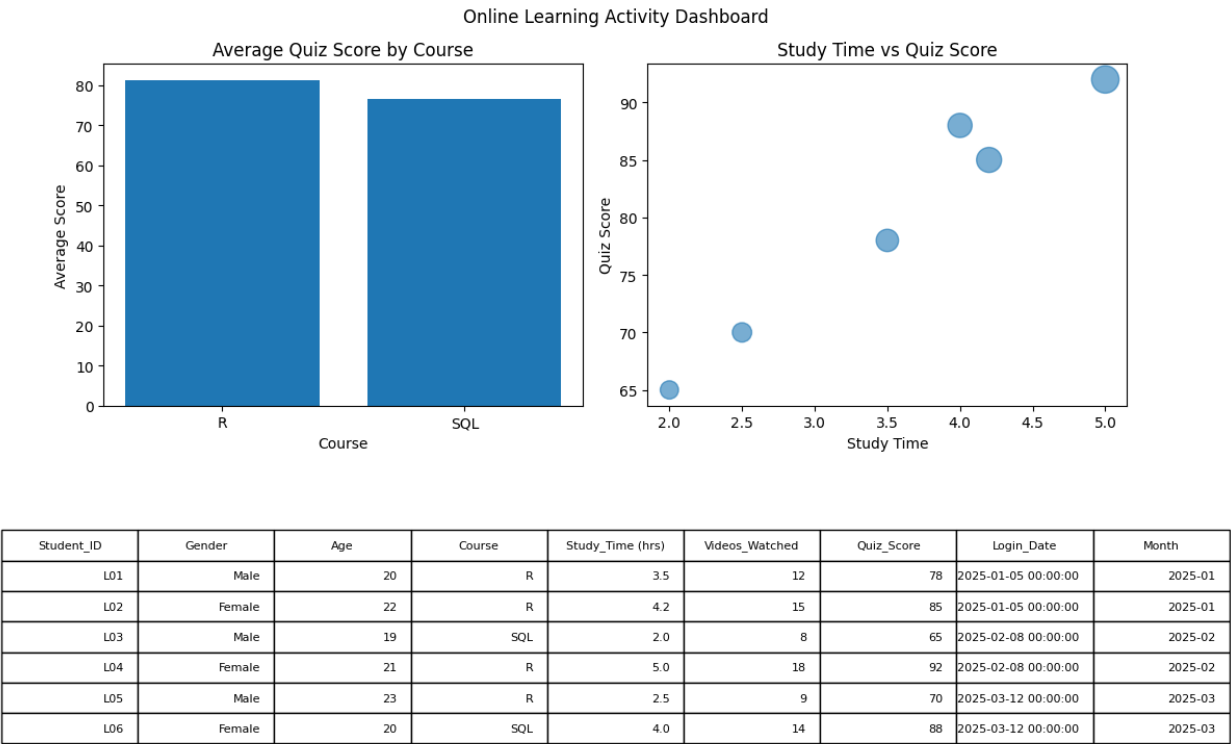
```
table.scale(1.2, 1.5)
```

```
plt.suptitle("Online Learning Activity Dashboard")
```

```
plt.tight_layout()
```

```
plt.show()
```

output:



Set 10 Survey Responses Analysis Dataset: Survey Responses
Survey ID 1 2 3 Question 1 A B Question 2 B Question 3 C A
C 1. A D B Create a grouped bar chart to visualize the
distribution of answers for Question 1. Label the chart
elements. 2. Generate a stacked bar chart to represent the

overall distribution of responses for all three questions. 3. 4. Build a table to show the survey response data for each survey. Label the table elements. Develop a Tableau dashboard combining the grouped bar chart, stacked bar chart, and the table for interactive exploration of survey responses.

A)

```
import pandas as pd
import matplotlib.pyplot as plt

data = {
    "Survey ID": [1, 2, 3],
    "Question 1": ["A", "B", "A"],
    "Question 2": ["B", "C", "D"],
    "Question 3": ["C", "A", "B"]
}

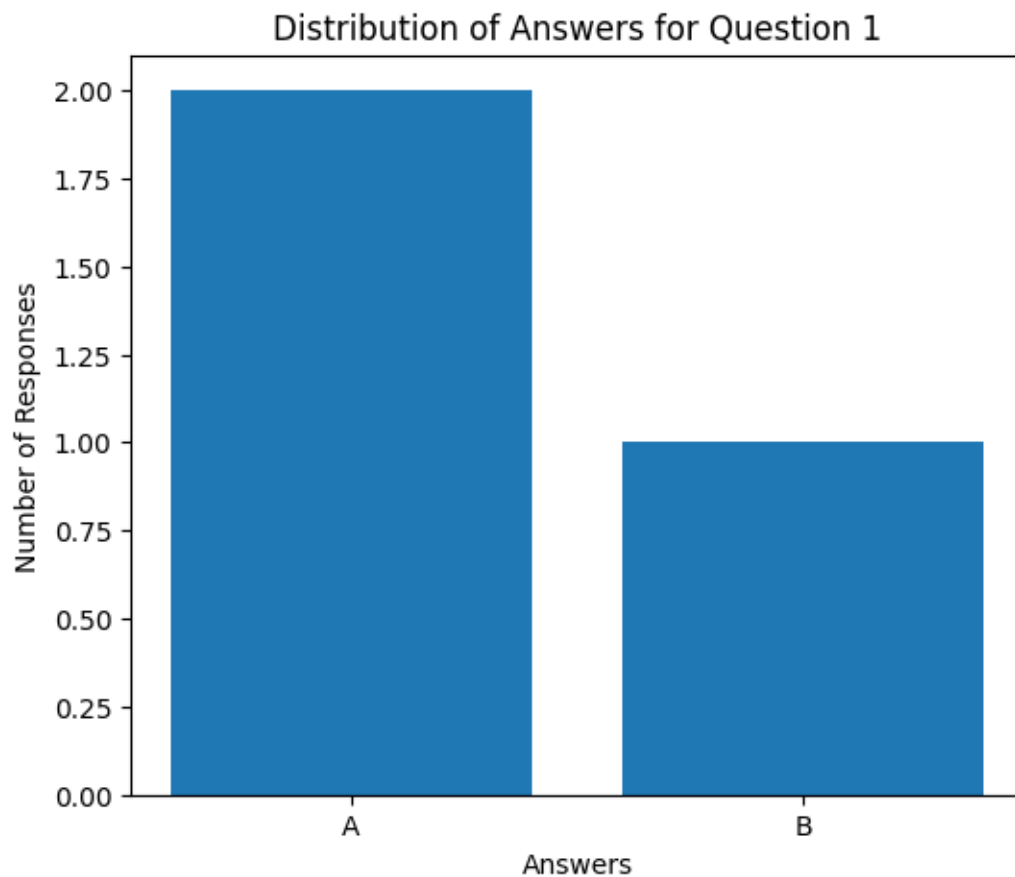
df = pd.DataFrame(data)
print(df)

import matplotlib.pyplot as plt

# Count responses
q1 = df["Question 1"].value_counts()

plt.figure(figsize=(6,5))
```

```
plt.bar(q1.index, q1.values)
plt.title("Distribution of Answers for Question 1")
plt.xlabel("Answers")
plt.ylabel("Number of Responses")
plt.show()
output:
```



b)

```
import pandas as pd
import matplotlib.pyplot as plt
```

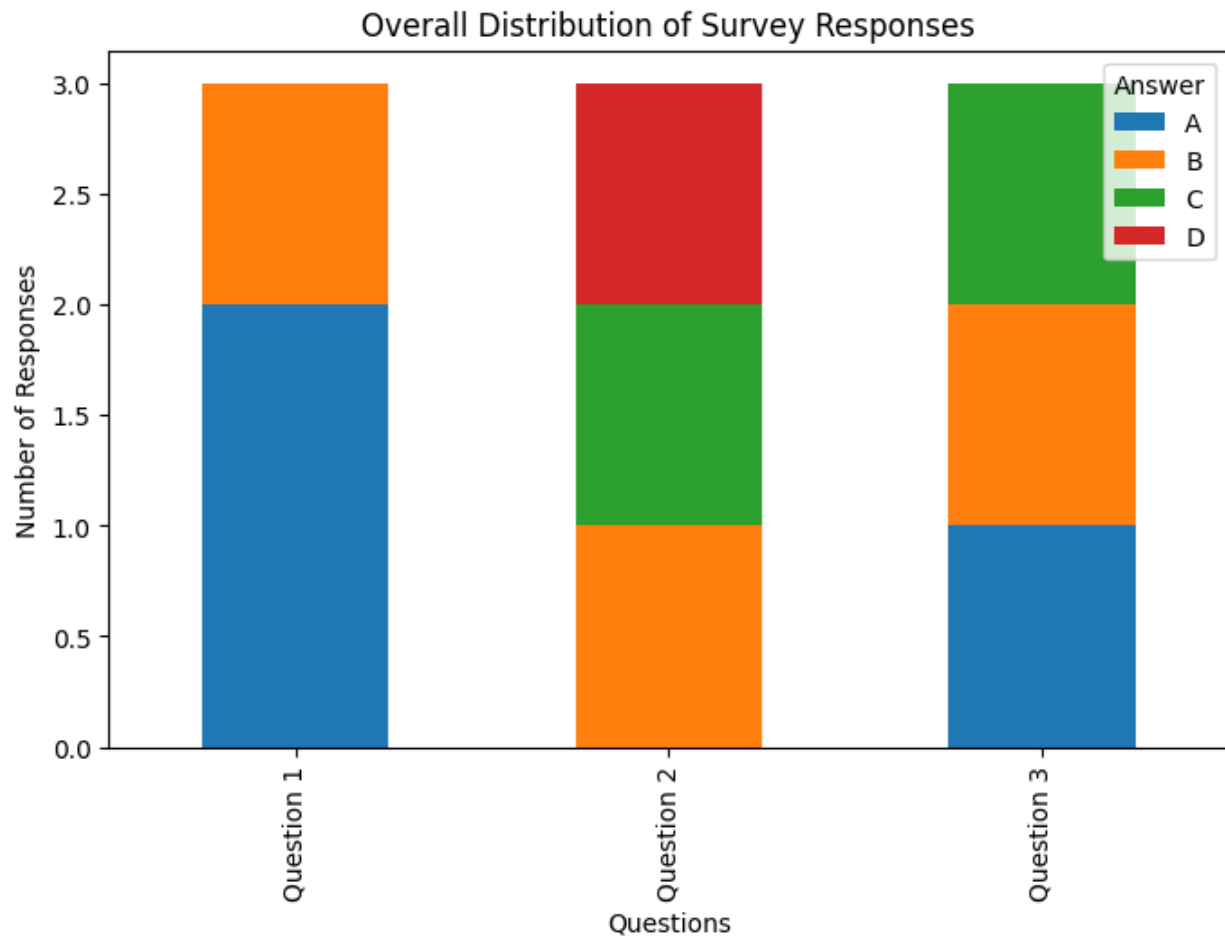
```
# Count each answer for every question
counts = pd.DataFrame({
    "Question 1": df["Question 1"].value_counts(),
    "Question 2": df["Question 2"].value_counts(),
    "Question 3": df["Question 3"].value_counts()
}).fillna(0)

counts.T.plot(
    kind="bar",
    stacked=True,
    figsize=(8,5)
)

plt.title("Overall Distribution of Survey Responses")
plt.xlabel("Questions")
plt.ylabel("Number of Responses")

plt.legend(title="Answer")

plt.show()
output:
```



c)

```
print("\nSurvey Response Table\n")
```

```
print(df.to_string(index=False))
```

output:

Survey Response Table

Survey ID	Question 1	Question 2	Question 3
-----------	------------	------------	------------

1	A	B	C
---	---	---	---

2	B	C	A
3	A	D	B

D)

```
import matplotlib.pyplot as plt
```

```
fig = plt.figure(figsize=(12,8))
```

```
# -----
```

```
# Grouped Bar Chart
```

```
# -----
```

```
ax1 = fig.add_subplot(221)
```

```
q1 = df["Question 1"].value_counts()
```

```
ax1.bar(q1.index, q1.values)
```

```
ax1.set_title("Question 1 Responses")
```

```
ax1.set_xlabel("Answer")
```

```
ax1.set_ylabel("Count")
```

```
# -----
```

```
# Stacked Bar Chart
```

```
# -----
```

```
ax2 = fig.add_subplot(222)
```

```
counts = pd.DataFrame({  
    "Question 1": df["Question 1"].value_counts(),  
    "Question 2": df["Question 2"].value_counts(),  
    "Question 3": df["Question 3"].value_counts()  
}).fillna(0)
```

```
counts.T.plot(kind="bar", stacked=True, ax=ax2)
```

```
ax2.set_title("Survey Response Distribution")
```

```
ax2.set_xlabel("Questions")
```

```
ax2.set_ylabel("Responses")
```

```
# -----
```

```
# Table
```

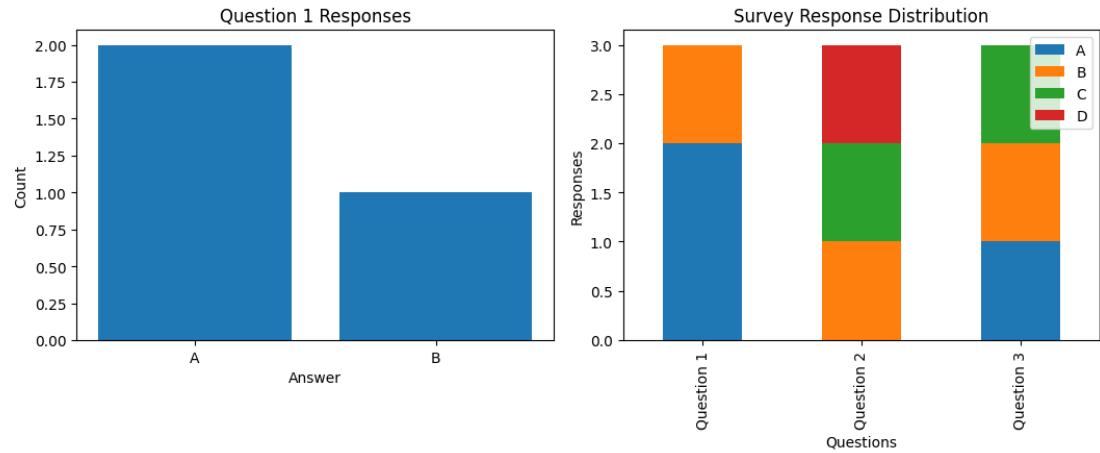
```
# -----
```

```
ax3 = fig.add_subplot(212)
```

```
ax3.axis("off")
```

```
table = ax3.table(  
    cellText=df.values,  
    colLabels=df.columns,  
    loc="center"  
)  
table.auto_set_font_size(False)  
table.set_fontsize(10)  
table.scale(1.2, 2)  
plt.suptitle("Survey Responses Dashboard")  
plt.tight_layout()  
plt.show()  
output:
```

Survey Responses Dashboard



Survey ID	Question 1	Question 2	Question 3
1	A	B	C
2	B	C	A
3	A	D	B

